Shaw Campbell

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Education

University of Cape Town — Cape Town, South Africa
BSc(Eng) Honours in Electrical & Computer Engineering — *2021 – 2025*

- Graduated with Honours; overall GPA 71%.
- Honours thesis: deep-learning deformable registration of ultrasound to MRI for prostate imaging (PyTorch, VoxelMorph, ANTs); semi-supervised training outperformed unsupervised, with affine pre-alignment and augmentation critical given limited data.

Experience

PSG Financial Services — Cape Town, South Africa
Graduate Software Developer (Intern) — *2026 – Present*

- Build and maintain production web applications (.NET / Angular, TypeScript) on an advice platform used by thousands of financial advisors; write C# end-to-end tests with Playwright.
- Migrated 100+ automated tests from a legacy framework to Playwright using AI coding tools (DeepSeek, OpenCode), cutting regression cycles from months to weeks.

Peralex Electronics — Cape Town, South Africa
Software Engineering Intern — *2025 (5 weeks)*

- Designed and built a Python desktop app for visualising openEMS electromagnetic simulation results across multiple antenna types, within a full Agile cycle.

Electrum Payments — Cape Town, South Africa
Software Development Intern — *2024 (4 weeks)*

- Delivered a feature end-to-end on a live production codebase (Java, Spring Boot, REST APIs) embedded in the engineering team; received a full-time return offer.

University of Cape Town — Cape Town, South Africa
Tutor & Laboratory Assistant, Electrical Engineering & Computer Science — *2023 – 2025*

- Tutored Signals & Systems I/II and introductory Python/Java programming to classes of 300+ students; Control Systems Laboratory Assistant (2023, 2024).

Selected Projects

Multi-Modal Deformable Image Registration — 2026

- Converting messy thesis Jupyter notebooks into a clean, modular, documented Python repository for training image-registration models.

TaskFlow: CPU-Inspired Task Scheduler (C++) — 2025

- Translated a full UML design into a working task scheduler implementing classical CPU scheduling algorithms and OOP design patterns, with CMake.

Honey Badger Predator Deterrent System (EEE4114F) — 2024

- Developed the networking and UI for an ESP32-based predator deterrence system for a penguin conservation project with BirdLife South Africa.

Extracurricular & Awards

IT & Maintenance Representative, Leo Marquard Hall House Committee — 2023

- Served on a committee overseeing a residence of 400+ students and hundreds of thousands of rands in funds; helped secure winter heater funding for residents.

Deans Merit List Award — 2022 – 2023

UCT Plus Gold Award (for over 60 hours of elected leadership position) — 2023

PSG Financial Services Merit Based Bursary (Full-ride two year bursary) — 2024 – 2025

Skills

Languages: Python, Java, C++, C, C#, TypeScript/JavaScript, HTML/CSS
Tools & Frameworks: TensorFlow, .NET, ASP.NET, Angular, Spring Boot, Playwright, Jupyter, Git, Linux, MS SQL Server, NUnit, Selenium